Distributionally Robust LMI Control

Direct Data-Driven and DRC Synthesis

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Chapter 1

Distributionally Robust LMI Synthesis for LTI Systems

1.1 Problem Setting [1]

Output Feedback Control.

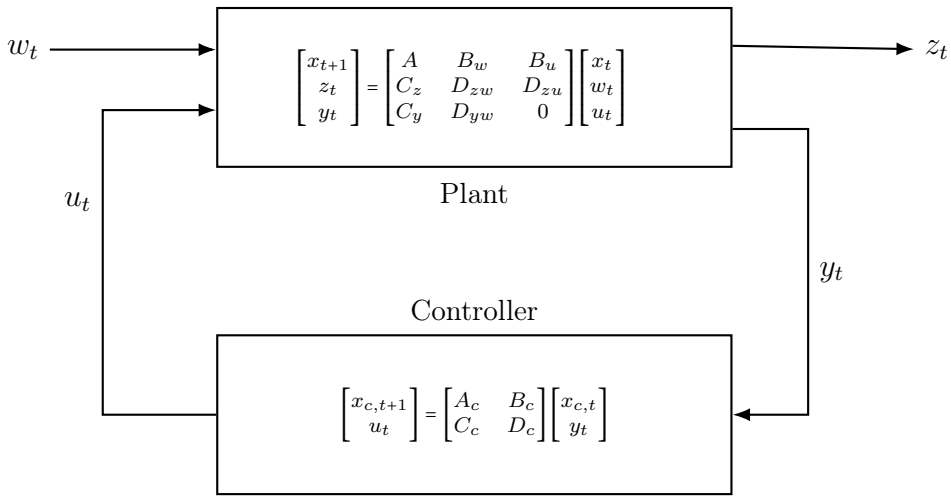


Figure 1.1: Closed-loop system with y feedback.

Define the combined state $\chi_t = [x_t^\top, x_{c,t}^\top]^\top$ and the closed-loop matrices

$$\mathcal{A} = \begin{bmatrix} A + B_u D_c C_y & B_u C_c \\ B_c C_y & A_c \end{bmatrix}, \quad \mathcal{B} = \begin{bmatrix} B_w + B_u D_c D_{yw} \\ B_c D_{yw} \end{bmatrix},$$
$$\mathcal{C} = \begin{bmatrix} C_z + D_{zu} D_c C_y & D_{zu} C_c \end{bmatrix}, \quad \mathcal{D} = D_{zw} + D_{zu} D_c D_{yw}.$$

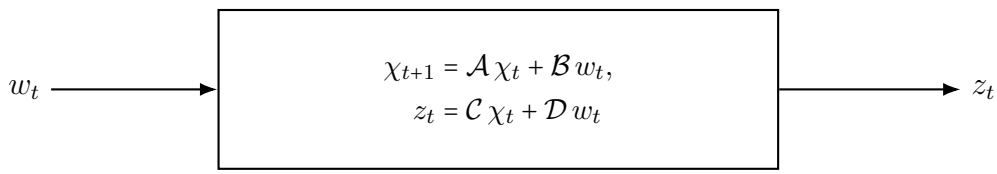


Figure 1.2: Lifted closed loop with state χ_t and matrices $(\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D})$.

The disturbance formulation is formulated in Appendix A.

1.2 Paper-like Model Setting

The *paper-like* configuration [2] refers to a linear time-invariant (LTI) system of the form:

$$\begin{bmatrix} x_{t+1} \\ z_t \\ y_t \end{bmatrix} = \begin{bmatrix} A & B_w & B_u \\ C_z & D_{zw} & D_{zu} \\ C_y & D_{yw} & 0 \end{bmatrix} \begin{bmatrix} x_t \\ w_t \\ u_t \end{bmatrix},$$

where $x_t \in \mathbb{R}^{n_x}$ denotes the state, $u_t \in \mathbb{R}^{n_u}$ the control input, and $w_t \in \mathbb{R}^{n_w}$ an exogenous disturbance. The vectors $y_t \in \mathbb{R}^{n_y}$ and $z_t \in \mathbb{R}^{n_z}$ represent the measured and performance outputs, respectively.

The continuous-time version of this model describes a flexible turbine–shaft–pitch system:

$$\dot{x}(t) = A_{CT}x(t) + B_{CT}u(t) + E_{CT}w(t),$$

with

$$A_{CT} = \begin{bmatrix} \frac{k_\omega}{J} & \frac{k_h}{J} & 0 & \frac{k_\phi}{J} & 0 & \frac{k_\beta}{J} & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ \frac{f_\omega}{m} & -\omega_f^2 & -2\zeta_f\omega_f & 0 & 0 & \frac{f_\beta}{m} & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ \frac{m_\omega}{I_t} & 0 & 0 & -\omega_t^2 & -2\zeta_t\omega_t & \frac{m_\beta}{I_t} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & -\omega_p^2 & -2\zeta_p\omega_p \end{bmatrix},$$

$$B_{CT} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \omega_p^2 \end{bmatrix}, \quad E_{CT} = \begin{bmatrix} \frac{k_v}{J} & 0 \\ 0 & 0 \\ \frac{f_v}{m} & \frac{f_v}{m} \\ 0 & 0 \\ \frac{m_v}{I_t} & \frac{m_v}{I_t} \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

which is discretized with sampling time T_s via

$$A = e^{A_{CT}T_s}, \qquad B_u = \int_0^{T_s} e^{A_{CT}\tau} B_{CT}\, d\tau, \qquad B_w = \int_0^{T_s} e^{A_{CT}\tau} E_{CT}\, d\tau.$$

The output matrices $(C_z, D_{zw}, D_{zu}, C_y, D_{yw})$ are selected to reflect a standard $\mathcal{H}_2$-type performance objective:

$$z_t = \begin{bmatrix} Q_x^{1/2} & 0 \\ 0 & R_u^{1/2} \end{bmatrix} \begin{bmatrix} x_t \\ u_t \end{bmatrix}, \qquad Q_x \succeq 0, R_u \succ 0 \qquad y_t = C_y x_t + D_{yw} w_t,$$

with $Q_x \succeq 0, R_u \succ 0$ and $D_{zw} = 0$.

1.3 Optimisation Problems [1]

Baseline

$$\min_{\theta \leftrightarrow (A_c, B_c, C_c, D_c)} J(\theta), \qquad \text{s.t. } \mathcal{A} \quad \text{isstable}$$

$$J(\theta) = \sup_{w \in \mathcal{W}} \quad \limsup_{T \to \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \, \|z_t\|^2 \tag{P1}$$

$$\text{s.t.} \quad \begin{aligned} \chi_{t+1} &= \mathcal{A} \chi_t + \mathcal{B} w_t, \\ z_t &= \mathcal{C} \chi_t + \mathcal{D} w_t \end{aligned}, \quad \chi_0 = 0$$

LMI

Introducing decision variables

$$\lambda \geq 0, \quad Q, \quad X = X^\top, \; Y = Y^\top, \; K, L, M, N,$$

and defining

$$\mathbb{P} = \begin{bmatrix} Y & I \\ I & X \end{bmatrix}, \qquad \mathbf{A} = \begin{bmatrix} AY + B_u M & A + B_u N C_y \\ K & XA + LC_y \end{bmatrix},$$

$$\mathbf{B} = \begin{bmatrix} B_w + B_u N D_{yw} \\ XB_w + LD_{yw} \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} C_z Y + D_{zu} M & C_z + D_{zu} N C_y \end{bmatrix}, \quad \mathbf{D} = D_{zw} + D_{zu} N D_{yw}.$$

The objective bacame:

$$\min_{\lambda \geq 0, \; Q, \; X = X^\top, \; Y = Y^\top, \; K, L, M, N} \quad \operatorname{trace}(Q \Sigma_{\text{nom}}) + \lambda \gamma^2 \tag{P2}$$

subject to a block LMI that depends on the ambiguity model.

Correlated model ($\mathcal{W}_{\text{cor}}$). With identity blocks I_w, I_z of matching sizes, enforce

$$\begin{bmatrix} -\mathbb{P} & 0 & 0 & \mathbf{A}^\top & \mathbf{C}^\top \\ 0 & -\lambda I_w & \lambda I_w & \mathbf{B}^\top & \mathbf{D}^\top \\ 0 & \lambda I_w & -Q - \lambda I_w & 0 & 0 \\ \mathbf{A} & \mathbf{B} & 0 & -\mathbb{P} & 0 \\ \mathbf{C} & \mathbf{D} & 0 & 0 & -I_z \end{bmatrix} \prec 0, \quad \mathbb{P} \succeq 0.$$

Independent model ($\mathcal{W}_{\text{ind}}$). Impose the pair of LMIs

$$\begin{bmatrix} -\mathbb{P} & \mathbf{A}^\top & \mathbf{C}^\top \\ \mathbf{A} & -\mathbb{P} & 0 \\ \mathbf{C} & 0 & -I_z \end{bmatrix} \prec 0, \qquad \begin{bmatrix} -\lambda I_w & \lambda I_w & \mathbf{B}^\top & \mathbf{D}^\top \\ \lambda I_w & -Q - \lambda I_w & 0 & 0 \\ \mathbf{B} & 0 & -\mathbb{P} & 0 \\ \mathbf{D} & 0 & 0 & -I_z \end{bmatrix} \prec 0, \quad \mathbb{P} \succeq 0.$$

Chapter 2

Direct Data-Driven Approach

2.1 Direct Data-Driven DRO–LMI via Young-Type Relaxation

Idea. We first estimate the nominal plant maps from one persistently exciting trajectory $\{x_k, u_k, y_k, z_k\}_{k=0}^{T}$ by one–step regression. The remaining data mismatch is treated as an *uncertainty budget* and injected into the robust LMI through a Young-inequality relaxation. This keeps the synthesis convex while avoiding an explicit probabilistic model of the identification errors.

One–step estimates and residual decomposition

Form the stacked data matrices

$$X_- = \begin{bmatrix} x_0 & \cdots & x_{T-1} \end{bmatrix}, \qquad U_- = \begin{bmatrix} u_0 & \cdots & u_{T-1} \end{bmatrix}, \qquad X_+ = \begin{bmatrix} x_1 & \cdots & x_T \end{bmatrix}.$$

$$Y_- = \begin{bmatrix} y_0 & \cdots & y_{T-1} \end{bmatrix}, \qquad Z_- = \begin{bmatrix} z_0 & \cdots & z_{T-1} \end{bmatrix},$$

Dataset preprocessing. Rather than relying on a single open-loop experiment, multiple trajectories $\{\Gamma_i\}_{i=1}^{N=50}$ are collected, each containing matrices

$$\Gamma_i = \{X_i, U_i, Y_i, Z_i, X_{i,+}\}, \qquad X_i \in \mathbb{R}^{n_x \times T_i},\ U_i \in \mathbb{R}^{n_u \times T_i},\ Y_i \in \mathbb{R}^{n_y \times T_i},\ Z_i \in \mathbb{R}^{n_z \times T_i},$$

all truncated to a common horizon $T = \min_i T_i$. Each dataset is initialized with random states (i.e. $x_0 \sim \mathcal{N}(0, I)$) to ensure excitation diversity, and only trajectories satisfying the persistent–excitation condition $\sigma_{\min}\left(\frac{1}{T}\Phi\Phi^\top\right) > \varepsilon_{\mathrm{PE}}$ are kept before medoid selection.

Before selecting a nominal dataset, the trajectories are *normalized* to prevent one signal from dominating the others. For each variable $k \in \{X, U, Y, Z, X_+\}$, a scale factor s_k is computed as the mean Frobenius norm across seeds,

$$s_k = \frac{1}{N} \sum_{i=1}^{N} \|\Gamma_i[k]\|_F + \varepsilon,$$

and all data are rescaled as $\tilde{\Gamma}_i[k] = \Gamma_i[k]/s_k$. This ensures that signals with different physical units contribute comparably to the distance metric.

Among the N realizations, the most representative one (the *medoid*) is selected as

$$i^\star = \arg\min_i \sum_k w_k \sum_{j=1}^{N} \left\|\tilde{\Gamma}_j[k] - \tilde{\Gamma}_i[k]\right\|_F^2,$$

where w_k are optional weighting coefficients. The final matrices used for identification are then taken from this single dataset,

$$(X_-, U_-, Y_-, Z_-, X_+) = (X_{i^\star}, U_{i^\star}, Y_{i^\star}, Z_{i^\star}, X_{i^\star,+}).$$

Motivation. A naive arithmetic average of the trajectories, e.g. $\bar{X} = \frac{1}{N}\sum_i X_i$, is undesirable because opposite sign components can cancel, artificially shrinking the state amplitudes and destroying the dynamical consistency between X_- and X_+. The medoid procedure avoids this by selecting one real, physically consistent trajectory that is closest (in the Frobenius sense) to all others, effectively reducing the influence of measurement noise and random variations without corrupting the system structure.

SysID Define $D_x = \begin{bmatrix} X_- \\ U_- \end{bmatrix}$ and use the (regularized) right inverse $D_x^\dagger := D_x^\top (D_x D_x^\top + \rho I)^{-1}$ (with $\rho \geq 0$, $\rho = 10^{-6}$ in our implementation). The nominal state update is

$$\underbrace{\begin{bmatrix} \hat{A} & \hat{B}_u \end{bmatrix}}_{\hat{O}_x} = X_+ D_x^\dagger, \qquad \hat{A} \in \mathbb{R}^{n\times n},\ \hat{B}_u \in \mathbb{R}^{n\times m}.$$

Let the *state residuals* be

$$R := X_+ - \hat{A}X_- - \hat{B}_u U_- \in \mathbb{R}^{n\times T}.$$

We extract a low–rank disturbance channel by principal directions of the residual covariance

$$S_R := \frac{1}{T} RR^\top, \quad \tilde{S}_R := \tfrac{1}{2}\big(S_R + S_R^\top\big) + \varepsilon I_n = U_s \operatorname{diag}(s_1, \ldots, s_n)\, U_s^\top,$$

and choose the minimal n_w such that $n_w := \min\left\{k : \sum_{i=1}^k s_i \ \geq\ \eta \sum_{i=1}^n s_i\right\}$.. Set

$$\hat{B}_w \widehat{\Sigma}_w^{1/2} := U_s(:, 1{:}n_w) \operatorname{diag}\big(\sqrt{s_1}, \ldots, \sqrt{s_{n_w}}\big), \qquad \hat{W} := \hat{B}_w^\dagger R,$$

with $\widehat{\Sigma}_w := \operatorname{diag}(s_1, \ldots, s_{n_w})$, and define the nominal disturbance covariance $\Sigma_{\text{nom}} := \frac{1}{2}\big(\frac{1}{T}\hat{W}\hat{W}^\top + (\frac{1}{T}\hat{W}\hat{W}^\top)^\top\big) + \varepsilon I_{n_w}$.

Define the empirical mean and covariance

$$\hat{\mu} := \frac{1}{T}\sum_{t=1}^{T} \hat{W}_t, \qquad \hat{\Sigma} := \frac{1}{T-1}\sum_{t=1}^{T} (\hat{W}_t - \hat{\mu})(\hat{W}_t - \hat{\mu})^\top. \tag{2.1}$$

The 2-Wasserstein distance between Gaussians decomposes as

$$W_2^2\big(\mathcal{N}(m_1, S_1), \mathcal{N}(m_2, S_2)\big) = \|m_1 - m_2\|_2^2 + \underbrace{\operatorname{tr}(S_1) + \operatorname{tr}(S_2) - 2\operatorname{tr}\Big(\big(S_2^{1/2} S_1 S_2^{1/2}\big)^{1/2}\Big)}_{=:\, d_B^2(S_1, S_2)}. \tag{2.2}$$

The plug-in estimator of the ambiguity radius is then

$$\gamma := d_B\big(\hat{\Sigma}, \Sigma_{\text{nom}}\big) = W_2\big(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \mathcal{N}(0, \Sigma_{\text{nom}})\big) \tag{2.3}$$

$$= \sqrt{\|\hat{\mu}\|_2^2 + \operatorname{tr}(\hat{\Sigma}) + \operatorname{tr}(\Sigma_{\text{nom}}) - 2\operatorname{tr}\Big(\big(\Sigma_{\text{nom}}^{1/2}\hat{\Sigma}\,\Sigma_{\text{nom}}^{1/2}\big)^{1/2}\Big)}. \tag{2.4}$$

Analogously, with $D_y = \begin{bmatrix} X_- \\ \hat{W} \end{bmatrix} \quad \Longrightarrow \quad \underbrace{\begin{bmatrix} \hat{C}_y & \hat{D}_{yw} \end{bmatrix}}_{\hat{O}_y} = Y_- D_y^\dagger,$

For the performance outputs we fix the weighting matrices as in the reference setup, defining $z = [\, Q_x^{1/2}x;\ R_u^{1/2}u \,]$ with

$$C_z = \begin{bmatrix} \operatorname{diag}(\sqrt{Q_x}) \\ 0_{n_u \times n_x} \end{bmatrix}, \quad D_{zu} = \begin{bmatrix} 0_{n_x \times n_u} \\ \operatorname{diag}(\sqrt{R_u}) \end{bmatrix}, \quad D_{zw} = 0, \quad Q_x = I_{n_x},\ R_u = I_{n_u}$$

so that the cost $\|z\|_2^2$ is consistent across the BL and DD formulations—this is a deliberate designer choice for fair comparison.

A data–driven radius for model error

We bound the *identification error* on the state map using the smallest nonzero singular value of D_x. Let $\sigma_{\min}$ be the smallest singular value of D_x, then

$$\|\Delta_x\|_F \le \beta \quad \text{with} \quad \beta := \frac{\|R\|_F}{\max(\sigma_{\min}, \epsilon)}.$$

proof:

$$\hat{O}_x = X_+ D^\dagger := (O_x D)D^\dagger + RD^\dagger = O_x + \Delta_x$$
$$\|\Delta_x\|_F \le \|RD^\dagger\|_F$$
$$\|D^\dagger\|_F \le \frac{1}{\sigma_{min}}$$

thus

$$\|\Delta_x\|_F \le \frac{\|R\|_F}{\max(\sigma_{\min}, \epsilon)}$$

Here Δ_x collects the stacked deviations in $[A\ B_u]$ relative to $\hat{O}_x$. This bound is conservative but computable directly from data.

$$\begin{bmatrix} A & B_u \end{bmatrix} = \underbrace{\begin{bmatrix} \hat{A} & \hat{B}_u \end{bmatrix}}_{\hat{O}_x} + \Delta_x, \quad \Delta_x = \begin{bmatrix} \Delta_A & \Delta_B \end{bmatrix}, \quad \|\Delta_i\|_F \le \beta_i$$

with

$$\beta_A = \beta\sqrt{\frac{n_x}{n_x + n_u}}, \quad \beta_B = \beta\sqrt{\frac{n_u}{n_x + n_u}}$$

Therephore, writing the perturbed lifted block as $\mathbf{A}_{\text{true}} = \mathbf{A}(\hat{A}, \hat{B}_u) + E_A + E_B$, the mixed terms $\mathbf{A}^\top P E_\cdot + E_\cdot^\top P \mathbf{A} = 2\langle E_\cdot, P\mathbf{A}\rangle$ are convexified by Young's inequality,

$$2\langle E_\cdot, Z\rangle \le s_\cdot \|E_\cdot\|_F^2 + \tfrac{1}{s_\cdot}\|Z\|_F^2, \quad Z = P\mathbf{A}, \quad s_\cdot > 0,$$

together with the data bounds $\|E_A\|_F \le \beta_A$ and $\|E_B\|_F \le \beta_B$ (induced by X, Y, M, N). These two pieces enter the LMI in complementary ways: (i) the scalar penalties $s_\cdot\beta_\cdot^2$ appear as a positive regularization of the Lyapunov corner, replacing the $(4,4)$ block by $-P + (\tau_A{+}\tau_B)I$ with $\left[\begin{smallmatrix} s_\cdot & \beta_\cdot \\ \beta_\cdot & \tau_\cdot \end{smallmatrix}\right] \succeq 0$; and (ii) the quadratic terms $(1/s_\cdot)\|P\mathbf{A}\|_F^2$ are encoded by augmenting the kernel with selector columns/rows and diagonal slack blocks $-s_A I$, $-s_B I$. The objective remains the Wasserstein DRO cost $\mathrm{tr}(Q\,\Sigma_{\mathrm{nom}}) + \lambda\gamma^2$ and the standard output–feedback change of variables $P = \left[\begin{smallmatrix} Y & I \\ I & X \end{smallmatrix}\right] \succ 0$ is retained. This yields a convex, data–consistent synthesis that is conservative (the second–order error term is upper–bounded) but implementable with LMIs. Formal derivations and the precise kernel substitutions are given in Appendix B.1; the post–optimization reconstruction of the nominal maps from the recovered errors, $A \leftarrow \hat{A} + \Delta_A$, $B_u \leftarrow \hat{B}_u + \Delta_B$, is detailed in Appendix B.2.

Final program. The Young-relaxed data-driven synthesis (correlated case) is

$$\min_{\substack{\lambda\ge 0,\ Q,\\ \{s_{A,i}\ge 0,\ \tau_{A,i}\ge 0\}_{i\in(1,n_x)},\\ s_B\ge 0,\ \tau_B\ge 0,\ \{t\}_{i\in(K,L,M,N)}\\ X=X^T,\ Y=Y^T,\ K,\ L,\ M,\ N}} \mathrm{tr}(Q\,\Sigma_{\mathrm{nom}}) + \lambda\gamma^2 + \sum_{j\in(A,B)}\sum_{i=1}^{n_x} \rho\left(s_{j,i} + \frac{\tau_{j,i}}{\beta_{j,i}^2+\varepsilon}\right) + \sum_{i\in(K,L,M,N)} \rho\left(t_i+\varepsilon\right)$$

$$\text{s.t. } P = \begin{bmatrix} Y & I \\ I & X \end{bmatrix} \succ 0, \quad \Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D})\Big|_{(4,4)} \leftarrow -P + \Big(\sum_{i=1}^{n_x}\tau_{A,i} + \tau_B\Big) I_{2n_x},$$

$$\begin{bmatrix} \Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D}) & \xi_A & \xi_B \\ \xi_A^\top & -\mathrm{Diag}(s_A) & 0 \\ \xi_B^\top & 0 & -s_B I_{n_x} \end{bmatrix} \prec 0,$$

$$\begin{bmatrix} s_{A,i} & \beta_{A,i} \\ \beta_{A,i} & \tau_{A,i} \end{bmatrix}\Big|_{i\in(1,n_x)} \succeq 0, \qquad \begin{bmatrix} s_B & \beta_B \\ \beta_B & \tau_B \end{bmatrix} \succeq 0, \quad \begin{bmatrix} t_j I_{n^j_{row}} & j \\ j^\top & t_j I_{n^j_{col}} \end{bmatrix}\Big|_{j\in(K,L,M,N)} \succeq 0.$$

where $\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D}$ are built from the nominal data-based maps $(\hat{A},\hat{B},\hat{C}_y,\hat{D}_{yw},\hat{C}_z,\hat{D}_{zu},\hat{D}_{zw})$. Analogously with the constraints in the *indipendent* disturbance case, with

$$\Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D})\Big|_{(2,2)} \leftarrow -P + \Big(\sum_{i=1}^{n_x}\tau_{A,i} + \tau_B\Big) I_{2n_x}.$$

2.2 Results: Young-Type Relaxation (Updated)

Setup

We compare the model-based design (BL) and the direct data-driven design (DD) under the *indipendent* (§1) disturbance setting, both synthesized with the Young-type DRO–LMI relaxation. The simulations reuse the same plant data and initial conditions used to generate the attached figures.

Objective vs empirical cost

The distributionally robust objective is

$$J_{\mathrm{BL}} = 14.3861, \qquad J_{\mathrm{DD}} = 16464.9354,$$

which favors BL. However, the *empirical* time–average stage cost satisfies

$$\overline{z^\top z}_{\mathrm{BL}} = 7.0992, \qquad \overline{z^\top z}_{\mathrm{DD}} = 4.0232,$$

indicating substantially lower observed operating cost for DD over the simulation horizon. This mismatch reflects the different risk measures: the DRO objective penalizes worst–case uncertainty (and is sensitive to stability margins), whereas the running average reports realized performance on the sampled trajectory.

Closed-loop stability

Discrete-time spectral radii of the closed-loop are

$$\rho(A_{\mathrm{cl,BL}}) = 0.99964 \approx 1.0, \qquad \rho(A_{\mathrm{cl,DD}}) = 1.00093 \approx 1.0.$$

BL is strictly inside the unit disk; DD slightly exceeds it, i.e., it is marginally unstable in the linear sense. This explains why the DRO objective penalizes DD more heavily despite its lower empirical cost.

Signal energies (RMS over the horizon)

Signal	BL (RMS)	DD (RMS)
x	0.9265	0.7066
x_c	88.4888	23.2676
u	0.0648	0.0094
y	1.3869	1.0656
z	0.8669	0.6609

DD reduces the RMS energy across all measured signals, including the performance output z and the control effort u, consistent with the lower time–average cost.

Controller and closed-loop norms

Controller and composite norms from the run:

$$\|B_c\|_F^{\mathrm{BL}} = 35.9714, \quad \|B_c\|_F^{\mathrm{DD}} = 195.0416,$$

$$\|A_{\mathrm{cl}}\|_2^{\mathrm{BL}} = 47.3826, \quad \|A_{\mathrm{cl}}\|_2^{\mathrm{DD}} = 197.4987,$$

with $\mathrm{cond}_2(A_{\mathrm{cl,DD}}) \approx 2.52 \times 10^9$. The larger gains and poorer conditioning for DD are consistent with the spectral-radius violation and the high DRO objective, despite the favorable empirical z-cost.

Discussion

Current evidence suggests a trade-off:

- **Empirical performance:** DD achieves lower running-average cost and lower RMS for z and u.

- **Robust objective & stability:** The DRO objective penalizes DD due to its marginal instability and larger gains; BL remains inside the unit disk with a much smaller J.

If the goal is to retain the empirical advantages of DD while restoring robust margins, one can tighten the Young penalties, add spectral-radius or Lyapunov slack constraints, or regularize controller norms more aggressively; this will likely increase the empirical cost toward BL but improve J and stability margins.

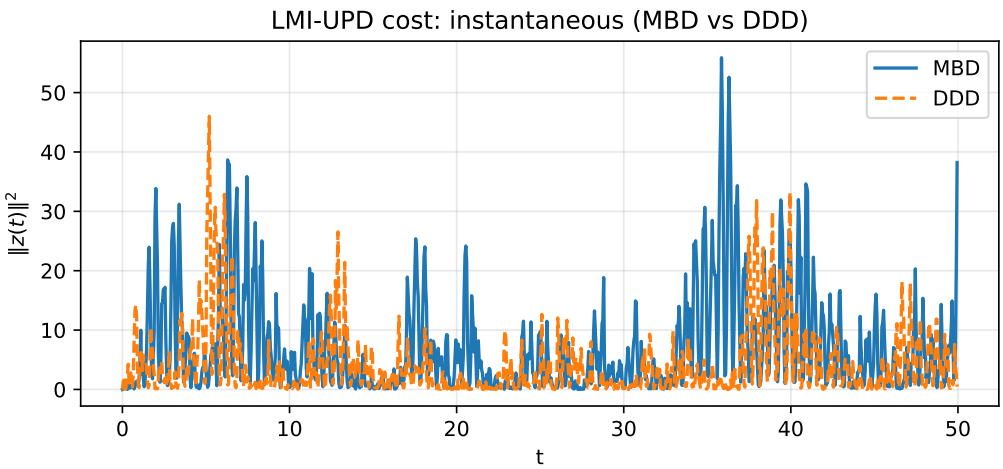


Figure 2.1: Instantaneous cost $z^\top z$: DD exhibits lower peaks along most of the trajectory.

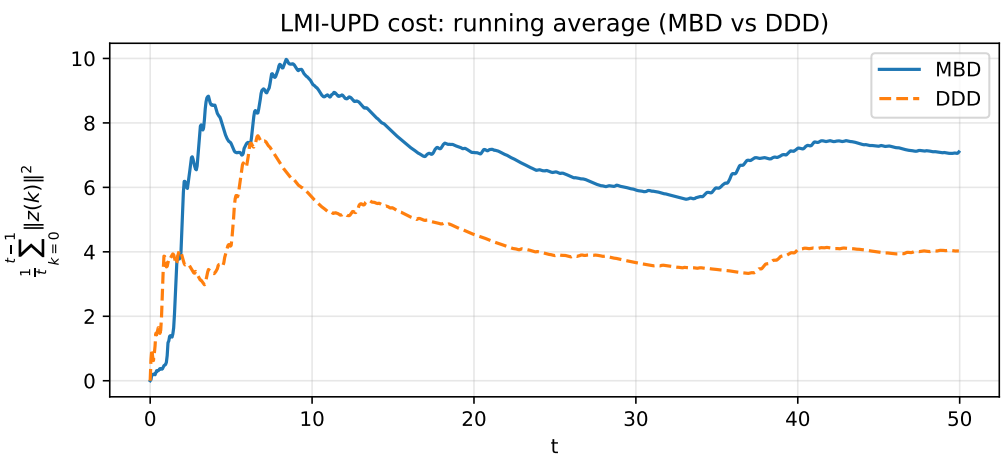


Figure 2.2: Running average cost: DD settles at a lower empirical level than BL.

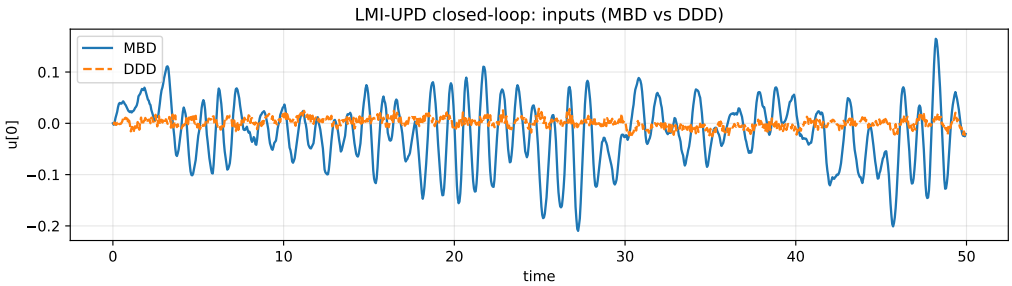


Figure 2.3: Control input u: DD requires substantially less control effort.

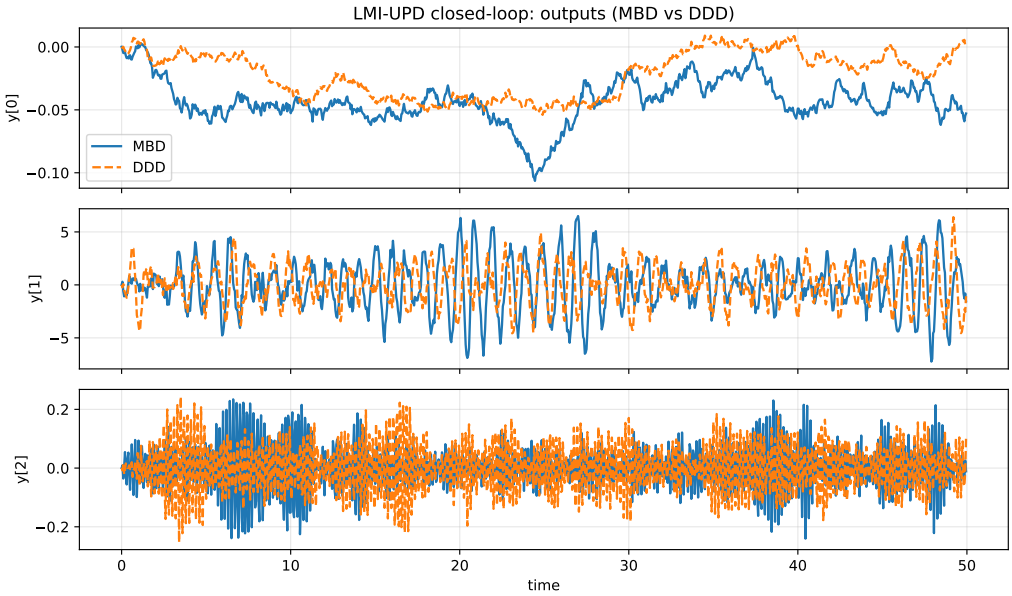


Figure 2.4: Measured outputs y: lower amplitude with DD, in line with reduced u.

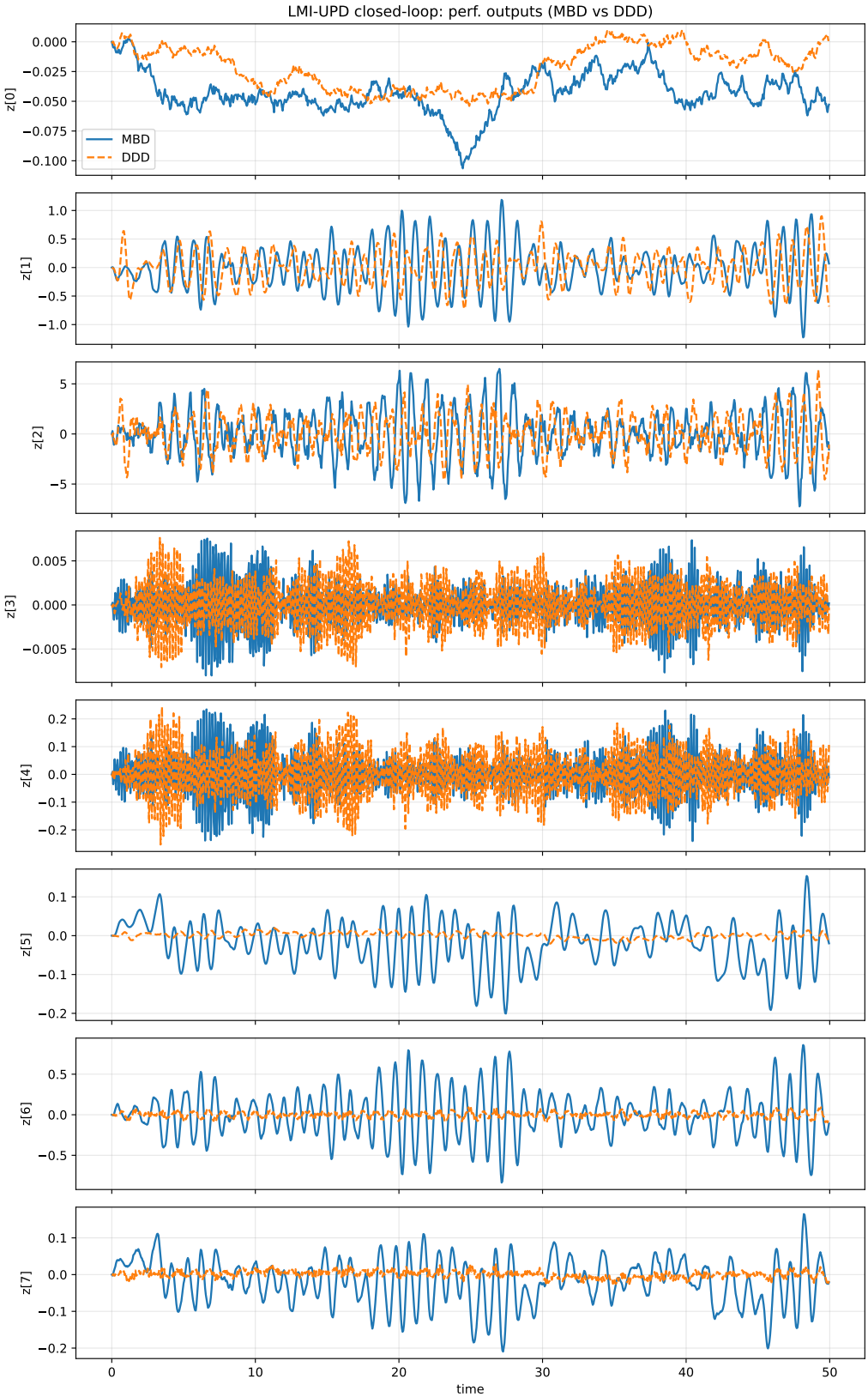


Figure 2.5: Performance outputs z: DD reduces error energy across components.

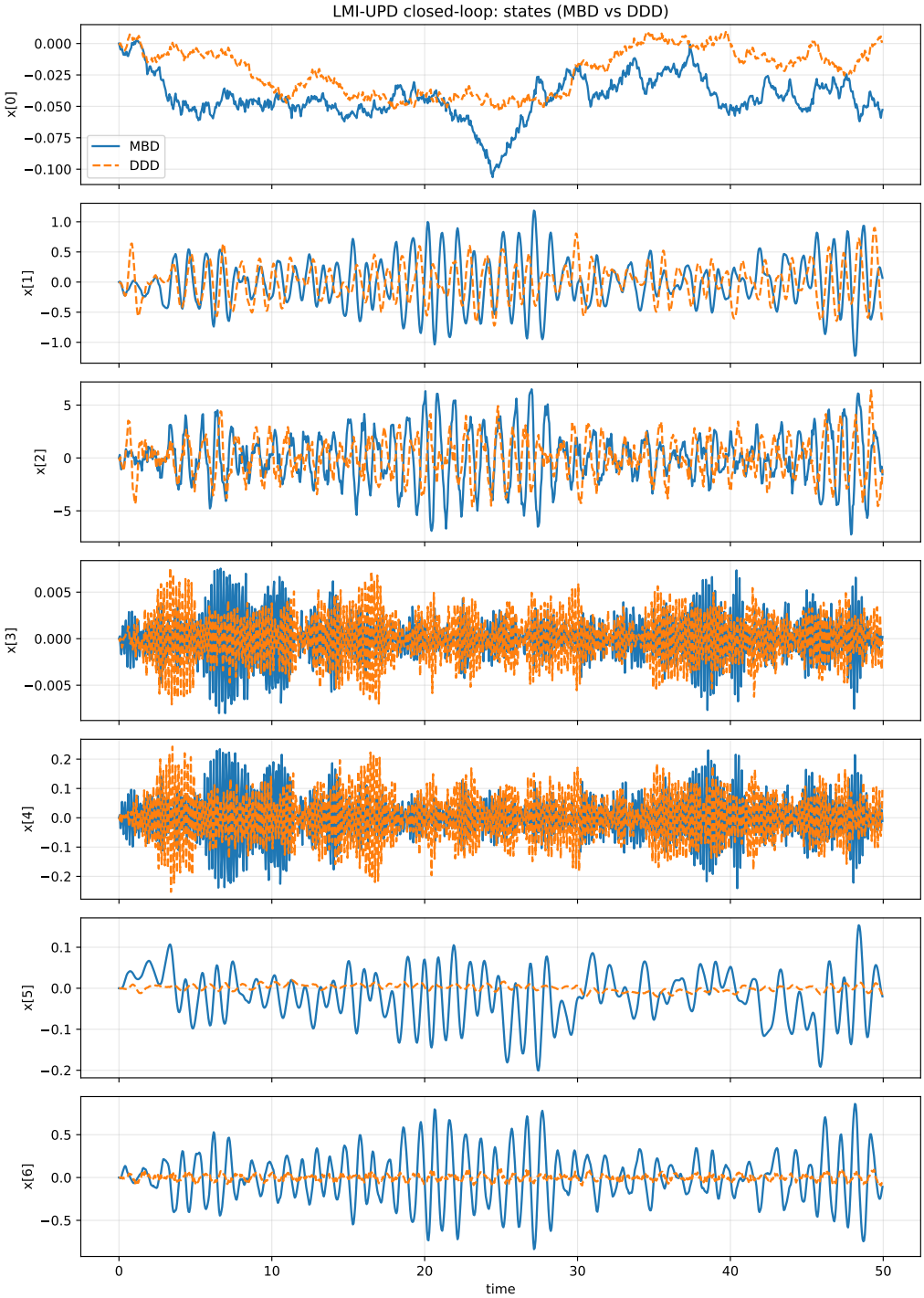


Figure 2.6: System states x: both stabilize the trajectory over the simulated window; DD shows lower RMS.

Appendix A

Disturbance Evaluation

The disturbance sequences w_t used in both the baseline and distributionally robust simulations are drawn from a *Wasserstein ambiguity set* [3, 4] centered at a nominal Gaussian model. The ambiguity set captures distributional uncertainty in the second-order statistics of the process noise, rather than assuming a fixed covariance.

Definition of the ambiguity set. For a given nominal covariance $\Sigma_{\text{nom}} \succ 0$ and Wasserstein radius $\gamma > 0$, the 2-Wasserstein ambiguity set is defined as

$$\mathcal{W}_2(\Sigma_{\text{nom}}, \gamma) = \left\{ \mathcal{N}(0, \Sigma) \;\middle|\; W_2(\mathcal{N}(0, \Sigma), \mathcal{N}(0, \Sigma_{\text{nom}})) \le \gamma \right\},$$

where $W_2(\cdot, \cdot)$ denotes the 2-Wasserstein distance between Gaussian measures:

$$W_2^2(\mathcal{N}(0, S_1), \mathcal{N}(0, S_2)) = \operatorname{tr}(S_1) + \operatorname{tr}(S_2) - 2 \operatorname{tr}\left[\left(S_2^{1/2} S_1 S_2^{1/2} \right)^{1/2} \right].$$

This set represents all Gaussian distributions whose covariance lies within distance γ (in the Bures metric) from the nominal covariance Σ_{nom}.

Sampling strategies. Two simulation modes are available:

1. **Independent (iid) disturbances:**

$$w_t \sim \mathcal{N}(0, \Sigma_{\text{nom}}), \quad t = 0, \ldots, T-1,$$

   corresponding to temporally uncorrelated samples inside the ambiguity ball. Implementation uses a Cholesky factorization $L = \operatorname{chol}(\Sigma_{\text{nom}})$ and generates $w_t = L\, z_t,\ z_t \sim \mathcal{N}(0, I)$.

2. **Correlated disturbances:** To allow temporal correlation while preserving the same marginal covariance, an autoregressive model of order one (AR(1)) is simulated:

$$w_{t+1} = \rho\, w_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, Q),$$

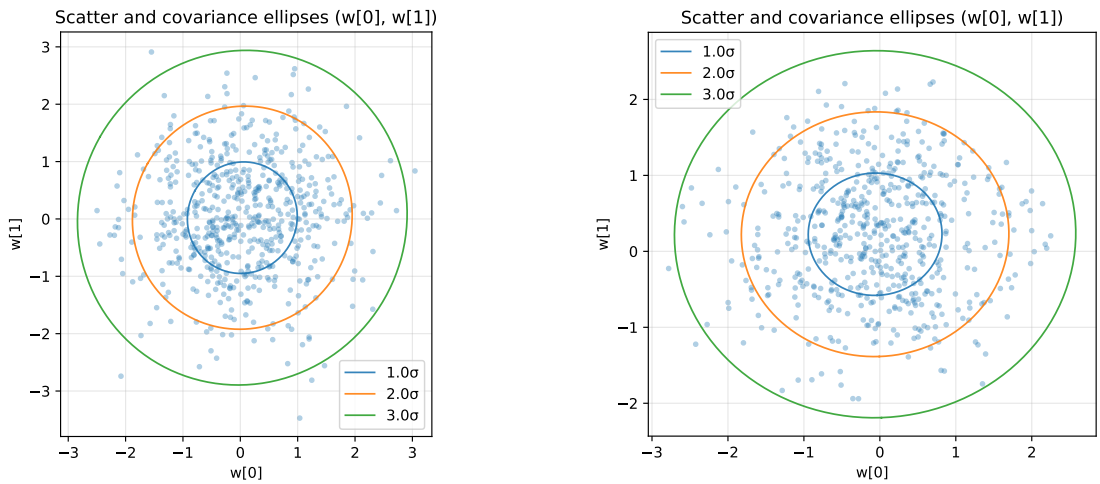
   with $0 < \rho < 1$ ensuring stability. The matrix Q is chosen so that

$$\Sigma_{\text{nom}} = \rho^2 \Sigma_{\text{nom}} + Q \quad \Rightarrow \quad Q = (1 - \rho^2)\Sigma_{\text{nom}}.$$

   This yields stationary correlated samples having marginal covariance Σ_{nom}, ensuring consistency with the ambiguity model.

In summary, the disturbances in simulation are not arbitrary noise sequences but samples from a *Wasserstein-consistent* uncertainty model. This maintains coherence between the data-driven identification, the LMI synthesis, and the subsequent Monte Carlo evaluation of closed-loop robustness.

Each figure below compares the *independent (iid)* and the *correlated (AR(1))* disturbance samplers that both lie in a 2-Wasserstein ball of radius γ around $\mathcal{N}(0, \Sigma_{\text{nom}})$. The iid and AR(1) samplers share the same stationary marginal $\mathcal{N}(0, \Sigma_{\text{nom}})$; they differ only in temporal structure (autocorrelation).



(a) Independent (iid): two-dimensional marginal samples (w_i, w_j) with nominal and empirical covariance ellipses. The empirical ellipse matches the nominal, confirming $\text{Cov}(w_t) = \Sigma_{\text{nom}}$ under iid sampling.

(b) Correlated (AR(1)): same marginal visualization. Marginals agree with Σ_{nom}; differences from iid are temporal, not marginal.

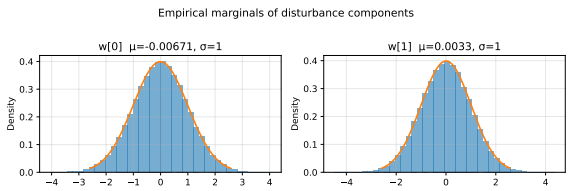
Figure A.1: Scatter and covariance ellipses. The panels visualize *marginal* covariance in the plane (w_i, w_j).

How iid vs correlated differ (and how they don't). Both modes satisfy $\mathbb{E}[w_t] = 0$ and $\text{Cov}(w_t) = \Sigma_{\text{nom}}$, hence the nearly identical scatter/ellipse and PDF panels. The difference is temporal:

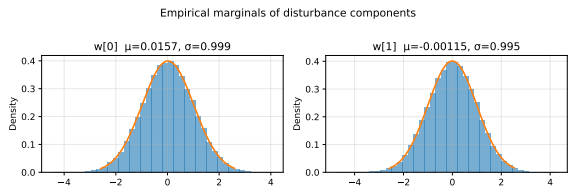
$$\text{iid:} \quad \text{Cov}(w_{t+k}, w_t) = 0 \ (k \neq 0), \qquad \text{AR(1):} \quad \text{Cov}(w_{t+k}, w_t) = \rho^{|k|}\Sigma_{\text{nom}}.$$

This is invisible in time-agnostic marginals. To expose it, add time-series diagnostics such as a trace of $w_i(t)$, an ACF plot (iid: spike at lag 0; AR(1): geometric decay ρ^k), or a lag-1 scatter (w_t, w_{t+1}) that aligns along the 45° line when correlated.

Technical caveat (boundary ellipses). The boundary covariances are constructed by eigenvalue shifts along principal axes of Σ_{nom}. This is exact in the commuting case (shared eigenvectors) and remains a useful visual surrogate otherwise; it illustrates the scale of admissible covariance perturbations at radius γ in the Bures/W_2 metric.

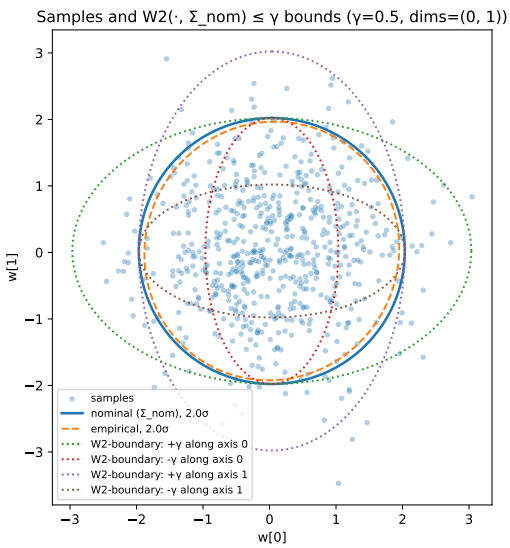


(a) Independent (iid): empirical marginal densities of disturbance components. Means are near zero and variances match the diagonal of Σ_{nom}.

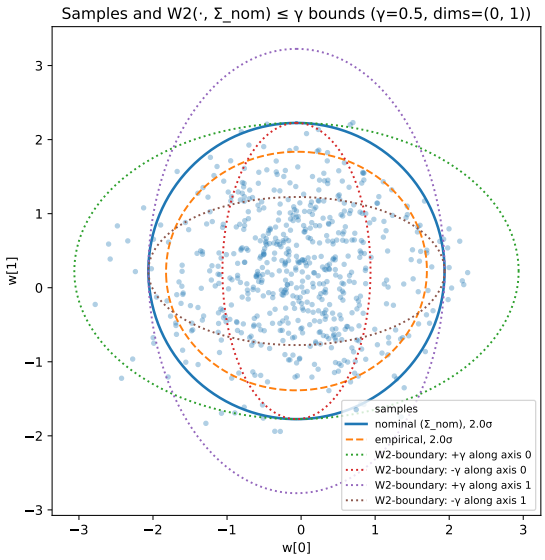


(b) Correlated (AR(1)): empirical marginal densities. As expected, they match the iid case; correlation does not alter stationary marginals.

Figure A.2: One-step marginal PDFs. Time order is ignored, so iid and AR(1) look nearly identical at the marginal level, as $T \to \inf$.



(a) Independent (iid): samples with nominal, empirical, and representative W_2-boundary ellipses. The nominal and empirical ellipses lie within the ball $W_2(\cdot, \Sigma_{\text{nom}}) \leq \gamma$.



(b) Correlated (AR(1)): correlated samples with the same W_2-boundary overlays. Correlation affects lag structure, not the W_2-marginal bounds.

Figure A.3: Wasserstein-ball overlays. Dotted ellipses illustrate covariances on the W_2 boundary around Σ_{nom}. They bound admissible *marginal* covariances, not individual points.

Appendix B

Dimonstrantion

B.1 Closed-loop lifting and decision variables

We keep the standard output-feedback change of variables

$$P = \begin{bmatrix} Y & I \\ I & X \end{bmatrix} \succ 0, \qquad \begin{aligned} \mathbf{A} &= \begin{bmatrix} \hat{A}Y + \hat{B}_u M & \hat{A} + \hat{B}_u N \hat{C}_y \\ K & X\hat{A} + L\hat{C}_y \end{bmatrix}, \\ \mathbf{B} &= \begin{bmatrix} \hat{B}_w + \hat{B}_u N \hat{D}_{yw} \\ X\hat{B}_w + L\hat{D}_{yw} \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} \hat{C}_z Y + \hat{D}_{zu} M & \hat{C}_z + \hat{D}_{zu} N \hat{C}_y \end{bmatrix}, \\ \mathbf{D} &= \hat{D}_{zw} + \hat{D}_{zu} N \hat{D}_{yw}. \end{aligned}$$

The decision set is $X = X^\top,\ Y = Y^\top,\ K,\ L,\ M,\ N,\ Q,\ \lambda \geq 0$ and the Wasserstein-DRO objective is

$$\min \quad \operatorname{tr}(Q\,\Sigma_{\text{nom}}) + \lambda\gamma^2.$$

Setup. Start from the correlated-noise kernel

$$\Xi(\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}) := \begin{bmatrix} -P & 0 & 0 & \mathbf{A}^\top & \mathbf{C}^\top \\ 0 & -\lambda I & \lambda I & \mathbf{B}^\top & \mathbf{D}^\top \\ 0 & \lambda I & -Q - \lambda I & 0 & 0 \\ \mathbf{A} & \mathbf{B} & 0 & -P & 0 \\ \mathbf{C} & \mathbf{D} & 0 & 0 & -I \end{bmatrix} \prec 0, \tag{$\star$}$$

with decision variables $P \succ 0$, $Q \succeq 0$, $\lambda \geq 0$, and where $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ are the usual lifted blocks built from the controller variables (X, Y, K, L, M, N) and the plant maps.

1. **Isolate the Lyapunov generator.** Focus on the $(1,4)$–$(4,1)$ corner that produces the Lyapunov term:

   $$M(\mathbf{A}, P) := \begin{bmatrix} -P & \mathbf{A}^\top \\ \mathbf{A} & -P \end{bmatrix} \prec 0.$$

   Starting from the Schur complement of the $(4,4)$ block $-P$ we obtain

   $$-P - \mathbf{A}^\top(-P)^{-1}\mathbf{A} = -P + \mathbf{A}^\top P^{-1}\mathbf{A} \prec 0. \tag{1}$$

   This form expresses discrete-time stability in the inverse-P coordinates.

To bring it to the classical Lyapunov form, apply a congruence transformation with $P^{1/2}$:

$$P^{1/2}(-P + \mathbf{A}^\top P^{-1}\mathbf{A})P^{1/2} \prec 0.$$

Expanding and introducing the scaled matrix

$$\tilde{\mathbf{A}} := P^{1/2}\mathbf{A}P^{-1/2},$$

we note that

$$\tilde{\mathbf{A}}^\top\tilde{\mathbf{A}} = (P^{-1/2}\mathbf{A}^\top P^{1/2})(P^{1/2}\mathbf{A}P^{-1/2}) = P^{-1/2}\mathbf{A}^\top P\mathbf{A}P^{-1/2},$$

so that

$$\mathbf{A}^\top P\mathbf{A} = P^{1/2}\tilde{\mathbf{A}}^\top\tilde{\mathbf{A}}P^{1/2}. \tag{2}$$

Substituting into (1) and simplifying gives

$$(P^{1/2}\mathbf{A}^\top P^{-1/2})(P^{1/2}\mathbf{A}P^{-1/2}) - I \prec 0 \quad\Longleftrightarrow\quad \tilde{\mathbf{A}}^\top\tilde{\mathbf{A}} - I \prec 0,$$

which is the normalized discrete-time stability condition. Undoing the normalization by multiplying on both sides by $P^{1/2}$ yields the standard Lyapunov inequality

$$\mathbf{A}^\top P\mathbf{A} - P \prec 0. \tag{L}$$

Hence the passage between the two expressions simply corresponds to expressing the same Lyapunov condition in two equivalent coordinate systems.

2. **Inject identification error.** Let $A = \hat{A} + \Delta_A$, $B_u = \hat{B}_u + \Delta_B$, with data-driven bounds $\|\Delta_A\|_F \le \beta_A$, $\|\Delta_B\|_F \le \beta_B$. In the lifted block, $\mathbf{A}$ depends *affinely* on (A, B_u), so replacing by $\hat{A}, \hat{B}_u$ plus errors gives

$$\mathbf{A}_{\text{true}} = \mathbf{A}(\hat{A}, \hat{B}_u) + E_A + E_B,$$

where E_A collects the contributions linear in Δ_A, and E_B those linear in Δ_B. In particular:

$$E_A = \begin{bmatrix} \Delta_A Y & \Delta_A \\ 0 & X\Delta_A \end{bmatrix} \qquad\qquad E_B = \begin{bmatrix} \Delta_B M & \Delta_B N C_y \\ 0 & 0 \end{bmatrix} \tag{B.1}$$

Expanding (L) with $\mathbf{A}_{\text{true}}$ gives

$$\mathbf{A}^\top P\mathbf{A} - P \;+\; \underbrace{\mathbf{A}^\top P(E_A{+}E_B) + (E_A{+}E_B)^\top P\mathbf{A}}_{\text{mixed terms}} \;+\; \underbrace{(E_A{+}E_B)^\top P(E_A{+}E_B)}_{\text{2nd order}} \prec 0,$$

where $\mathbf{A} := \mathbf{A}(\hat{A}, \hat{B}_u)$ is the nominal lifted block.

3. **Identify the mixed inner products.** The two mixed terms can be written as Frobenius inner products:

$$\mathbf{A}^\top P E_A + E_A^\top P\mathbf{A} = 2\langle E_A,\ Z_A\rangle, \qquad Z_A := P\mathbf{A},$$

$$\mathbf{A}^\top P E_B + E_B^\top P\mathbf{A} = 2\langle E_B,\ Z_B\rangle, \qquad Z_B := P\mathbf{A},$$

(here both couple to the same $P\mathbf{A}$ because E_A, E_B perturb the same lifted block). The second-order term is nonnegative and can be dropped for a conservative bound, or we could bound it with $\|P\|_2(\beta_A + \beta_B)^2 I_{2n_x}$ if tighter accounting is needed.

4. **Apply Young's inequality and encode it.**

Each mixed cross term $2\langle E_{A,i}, Z_{A,i}\rangle$ ($i \in \{A, B\}$) is non-convex, because it couples the uncertain matrix error $E_{A,i}$ (not a decision variable) and the matrix $Z_{A,i}$ (which depends affinely on the controller variables). To maintain convexity, we upper-bound it via Young's inequality:

$$2\langle E_{A,i}, Z_{A,i}\rangle \;\le\; s_{A,i}\|E_{A,i}\|_F^2 + \frac{1}{s_{A,i}}\|Z_{A,i}\|_F^2, \qquad s_{A,i} > 0.$$

Given the identification bound $\|E_{A,i}\|_F \le \beta_{A,i}$, this becomes

$$2\langle E_{A,i}, Z_{A,i}\rangle \;\le\; s_{A,i}\beta_{A,i}^2 + \frac{1}{s_{A,i}}\|Z_{A,i}\|_F^2, \qquad Z_{A,i} = P\mathbf{A}.$$

The first term $s_{A,i}\beta_{A,i}^2$ is a scalar uncertainty penalty; the second $\frac{1}{s_{A,i}}\|Z_{A,i}\|_F^2$ is a quadratic form in $Z_{A,i}$ that must be carried into the matrix inequality to maintain the convexity of the problem. Both pieces are encoded in convex form as follows.

Norm bounds. Using the Frobenius submultiplicativity $\|AXB\|_F \le \|A\|_2\|X\|_F\|B\|_2$ and the block norm identity $\left\|\begin{bmatrix} U & V \\ W & Z \end{bmatrix}\right\|_F^2 = \|U\|_F^2 + \|V\|_F^2 + \|W\|_F^2 + \|Z\|_F^2$, we obtain:

$$\begin{aligned} \|E_A\|_F^2 &= \|\Delta_A Y\|_F^2 + \|\Delta_A\|_F^2 + \|X\Delta_A\|_F^2 \\ &\le (\|Y\|_2^2 + 1 + \|X\|_2^2)\,\|\Delta_A\|_F^2, \end{aligned} \tag{B.2}$$

$$\begin{aligned} \|E_B\|_F^2 &= \|\Delta_B M\|_F^2 + \|\Delta_B N C_y\|_F^2 \\ &\le (\|M\|_2^2 + \|NC_y\|_2^2)\,\|\Delta_B\|_F^2. \end{aligned} \tag{B.3}$$

Therefore,

$$\boxed{\|E_A\|_F \le \sqrt{1 + \|X\|_2^2 + \|Y\|_2^2}\,\beta_A = \beta_A, \qquad \|E_B\|_F \le \sqrt{\|M\|_2^2 + \|NC_y\|_2^2}\,\beta_B = \beta_B.} \tag{B.4}$$

Consider those from **BL** simulation's results:

$$\|M\|_2 \approx 0.15, \quad \|N\|_2 \approx 0.6, \quad \|X\|_2 \approx 2.5 \times 10^5, \quad \|Y\|_2 \approx 1.0$$

(a) Scalar PSD blocks for the uncertainty radii. The scalar constraint $s_{A,i}\beta_{A,i}^2 \le s_{A,i}\tau_{A,i}$ is represented through the 2×2 positive-semidefinite (PSD) matrix

$$\begin{bmatrix} s_{A,i} & \beta_{A,i} \\ \beta_{A,i} & \tau_{A,i} \end{bmatrix} \succeq 0.$$

By the Schur complement lemma this is equivalent to $s_{A,i} > 0$ and $\tau_{A,i} \ge \beta_{A,i}^2/s_{A,i}$. It convexly enforces the product $s_{A,i}\tau_{A,i} \ge \beta_{A,i}^2$ without introducing the nonlinear reciprocal $1/s_{A,i}$.

For the AA cross term we use *directional* slacks

$$s_A \in \mathbb{R}_{\ge 0}^{n_x}, \qquad \tau_A \in \mathbb{R}_{\ge 0}^{n_x},$$

evaluated per eigendirection of the empirical residual covariance, i.e.

$$S^{\text{base}} = \tfrac{1}{T} RR^\top = U_A \operatorname{diag}(s)\, U_A^\top,$$

where s are the eigenvalues and U_A the corresponding eigenvectors. The directional weights are obtained as normalized square roots of the eigenvalues:

$$w_A^{(i)} = \frac{\sqrt{s_i}}{\max_j \sqrt{s_j}}, \qquad i = 1, \dots, n_x,$$

so that the direction with the largest residual variance has weight one. Using these weights, the selector is rotated as $S_A := S_A^{\text{base}} U_A$, and the per–direction uncertainty bounds are

$$\beta_A = (\beta_a\, w_A^{(i)})_{i=1}^{n_x}.$$

We impose for each $i = 1, \dots, n_x$:

$$\begin{bmatrix} s_{A,i} & \beta_{A,i} \\ \beta_{A,i} & \tau_{A,i} \end{bmatrix} \succeq 0, \quad \Longleftrightarrow \quad \tau_{A,i} \geq \beta_{A,i}^2 / s_{A,i}, \;\; s_{A,i} > 0.$$

For the AB cross term we keep a *scalar* slack $s_B \geq 0$ with companion $\tau_B \geq 0$ and

$$\begin{bmatrix} s_B & \beta_B \\ \beta_B & \tau_B \end{bmatrix} \succeq 0.$$

The quadratic terms $\|Z_{A,i}\|_F^2$ are then encoded via these diagonal (vector) and scalar slack blocks to preserve convexity.

(b) Augmentation of the main LMI for the $(1/s_{A,i})\|Z_{A,i}\|_F^2$ term. The second contribution, $\frac{1}{s_{A,i}}\|Z_{A,i}\|_F^2$, cannot appear as a scalar in the cost; it must enter the block inequality in the same location where the Lyapunov term $\mathbf{A}^\top P \mathbf{A} - P$ acts. To embed it linearly, we enlarge the main LMI using a Schur-complement trick: introduce an auxiliary matrix column $\xi_{A,i}$ and append a row/column pair corresponding to the slack variable $s_{A,i} > 0$:

$$\begin{bmatrix} \Xi(\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}) & \xi_A & \xi_B \\ \xi_A^\top & -\operatorname{Diag}(s_A) I & 0 \\ \xi_B^\top & 0 & -s_B I \end{bmatrix} \prec 0. \tag{†}$$

This extended block is equivalent (by Schur complement) to the condition

$$\Xi(\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}) - \frac{1}{s_A} \xi_A \xi_A^\top - \frac{1}{s_B} \xi_B \xi_B^\top \prec 0,$$

so the added rows/columns inject precisely the negative penalties $-\frac{1}{s_{A,i}} \xi_{A,i} \xi_{A,i}^\top$ that represent $+\frac{1}{s_{A,i}}\|Z_{A,i}\|_F^2$ in the quadratic form. Intuitively, the selectors $s_{A,i}$ identify which sub-block of the Lyapunov matrix is affected by the identification error. In practice we choose $\xi_A = \xi_B = \begin{bmatrix} I_{n_x} & 0 \end{bmatrix}^\top$, so that the perturbation acts on the "state" part of the $\Xi(\cdot)$ kernel corresponding to the $(-P)$ block.

Together, the scalar PSD constraints encode the bounded-error radius ($s_{A,i}\beta_{A,i}^2$ term), while the augmented LMI (†) embeds the $\frac{1}{s_{A,i}}\|P\mathbf{A}\|_F^2$ contribution directly

in the matrix inequality. This yields a convex relaxation of the non-convex cross-term $2\langle E_{A,i}, Z_{A,i}\rangle$, ensuring that all remaining constraints are affine or convex in the decision variables.

(c) Where the selector acts and why it is sparse. The cross-term generated by the A–uncertainty, $\mathbf{A}^\top P E_A + E_A^\top P \mathbf{A} = 2\langle E_A,\ P\mathbf{A}\rangle$, arises *only* in the state–state Lyapunov coupling of the kernel $\Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D})$, i.e., the $(4,4)$ block $(-P)$ and its $(1,4)$–$(4,1)$ off-diagonals in $\star$. Consequently, the Young penalty $\frac{1}{s_{A,i}}\|Z_{A,i}\|_F^2$ with $Z_{A,i} = P\mathbf{A}$ must be injected *at that same location.* We therefore augment the LMI by appending a column/row that *selects only the state coordinates* and all other entries of $\xi_{A,i}$ are set to zero because the A–perturbation does *not* generate cross-terms in the remaining rows/columns (output or disturbance couplings).

5. **Regularization term (convex surrogate).** The original penalty $\rho\left(s_{A,i} + s_{A,i}^{-1}\right)$ is replaced using $\tau_{A,i} \geq \beta_{A,i}^2/s_{A,i}$, which implies $s_{A,i}^{-1} \leq \tau_{A,i}/\beta_{A,i}^2$. Hence a convex upper bound is

$$\mathcal{R} = \rho \sum_{i=1}^{n_x}\left(s_{A,i} + \frac{\tau_{A,i}}{\beta_{A,i}^2}\right) + \rho\left(s_B + \frac{\tau_B}{\beta_B^2}\right),$$

optionally with a small $\varepsilon > 0$ for numerical safety:

$$\mathcal{R}_\varepsilon = \rho \sum_{i=1}^{n_x}\left(s_{A,i} + \frac{\tau_{A,i}}{\beta_{A,i}^2 + \varepsilon}\right) + \rho\left(s_B + \frac{\tau_B}{\beta_B^2 + \varepsilon}\right).$$

To control the operator norms of the learned matrices K, L, M, N, introduce auxiliary scalar variables $t_K, t_L, t_M, t_N \geq 0$ and impose the epigraph LMIs

$$\begin{bmatrix} t_K I_{n_x} & K \\ K^\top & t_K I_{n_x} \end{bmatrix} \succeq 0, \qquad \begin{bmatrix} t_L I_{n_x} & L \\ L^\top & t_L I_{n_y} \end{bmatrix} \succeq 0, \tag{B.5}$$

$$\begin{bmatrix} t_M I_{n_u} & M \\ M^\top & t_M I_{n_x} \end{bmatrix} \succeq 0, \qquad \begin{bmatrix} t_N I_{n_u} & N \\ N^\top & t_N I_{n_y} \end{bmatrix} \succeq 0. \tag{B.6}$$

These LMIs ensure $\|K\|_2 \leq t_K$, $\|L\|_2 \leq t_L$, $\|M\|_2 \leq t_M$, and $\|N\|_2 \leq t_N$.

A convex spectral-norm regularization term is then added to the objective:

$$\mathcal{R}_{\text{spec}} = \rho\,(t_K + t_L + t_M + t_N),$$

where the weights $\rho > 0$ tune the relative penalization of each block.

Optionally, a small numerical offset $\varepsilon > 0$ can be included to improve conditioning, yielding

$$\mathcal{R}_{\text{spec},\varepsilon} = \rho\,[(t_K + \varepsilon) + (t_L + \varepsilon) + (t_M + \varepsilon) + (t_N + \varepsilon)].$$

Together with the previous convex surrogate term $\mathcal{R}_\varepsilon$, these define the complete regularized optimization problem, with $\mathcal{R} = \mathcal{R}_\varepsilon + \mathcal{R}_{\text{spec},\varepsilon}$:

$$\min_{\text{decision vars}} \quad J_{\text{main}} + \mathcal{R} \quad \text{s.t.} \quad \text{LMI constraints above.}$$

6. **Substitution in the kernel.** In code, the Young penalties act on the state–state Lyapunov corner of $\Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D})$. Concretely, the $(4,4)$ block $(-P)$, which

corresponds to the Lyapunov generator in the Schur complement $-P + \mathbf{A}^\top P^{-1}\mathbf{A}$, is replaced by a regularized block that incorporates the scalar penalties produced by Young's inequality.

Starting from the bound

$$2\langle E_{A,i}, Z\rangle \;\le\; s_{A,i}\,\beta_{A,i}^2 + \frac{1}{s_{A,i}}\|Z\|_F^2, \qquad Z = P\mathbf{A},$$

each uncertainty component ($i \in \{A, B\}$) contributes a scalar term $s_{A,i}\beta_{A,i}^2$ and a quadratic term $\frac{1}{s_{A,i}}\|P\mathbf{A}\|_F^2$. The scalar term is encoded as a positive multiple of the identity on the same state–state block where $-P$ acts, through the auxiliary variables $\tau_{A,i}$ satisfying the Schur constraint $\begin{bmatrix} s_{A,i} & \beta_{A,i} \\ \beta_{A,i} & \tau_{A,i} \end{bmatrix} \succeq 0$, which enforces $\tau_{A,i} \ge \beta_{A,i}^2/s_{A,i}$.

Mathematically, the Lyapunov block is therefore augmented as

$$-P \quad\longrightarrow\quad -P + \sum_{i=1}^{n_x} \tau_{A,i} S_{A,i} S_{A,i}^\top + \tau_B S_B S_B^\top,$$

where ξ_A and ξ_B are the selector matrices acting on the state subspace. Since both selectors operate on the same coordinates, this sum collapses to

$$-P + \Big(\sum_{i=1}^{n_x} \tau_{A,i} + \tau_B\Big)\,\xi_{\text{state}},$$

with $\xi_{\text{state}} = \xi_A\xi_A^\top \simeq I_{2n_x}$. Replacing the projector by the identity yields a conservative but simpler formulation used in the implementation:

$$\texttt{state_blk} \;:=\; -P + \Big(\sum_{i=1}^{n_x} \tau_{A,i} + \tau_B\Big) I_{2n_x},$$

Finally, we append the selector columns/rows ξ_A, ξ_B with diagonal slacks $-s_A I$ and $-s_B I$ to capture the quadratic penalty $\frac{1}{s_{A,i}}\|P\mathbf{A}\|_F^2$, leading to the extended LMI

$$\begin{bmatrix} \Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D}) & \xi_A & \xi_B \\ \xi_A^\top & -s_A I & 0 \\ \xi_B^\top & 0 & -s_B I \end{bmatrix} \prec 0 \quad \text{with} \quad \Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D})\Big|_{(4,4)} \leftarrow \texttt{state_blk}.$$

By Schur complement this is equivalent to

$$\Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D}) - \frac{1}{s_A}\xi_A\xi_A^\top - \frac{1}{s_B}\xi_B\xi_B^\top \prec 0,$$

which is precisely the matrix form of the Young bounds for the mixed cross terms. The substitution of $(-P)$ by $\texttt{state_blk} = -P + (\tau_A + \tau_B)I$ thus represents the scalar regularization of the Lyapunov block induced by the bounded identification error.

B.2 Post-optimization recovery of Δ_A and Δ_B

Inputs. After solving the program, suppose we have optimal quantities

$$P^\star \succ 0, \quad X^\star = X^{\star\top}, \quad Y^\star = Y^{\star\top}, \quad M^\star,\ N^\star,\ s_A^\star > 0,\ s_B^\star > 0,\ \tau_A^\star,\ \tau_B^\star,$$

together with nominal identified maps $\hat{A}$, $\hat{B}_u$, $\hat{C}_y$ and data-driven radii β_A, β_B that induce the lifted bounds

$$\|E_A\|_F \le \beta_A := \sqrt{1 + \|X^\star\|_2^2 + \|Y^\star\|_2^2}\,\beta_A, \qquad \|E_B\|_F \le \beta_B := \sqrt{\|M^\star\|_2^2 + \|N^\star\|_2^2\|\hat{C}_y\|_2^2}\,\beta_B.$$

Step 1: Adversarial directions for the Young bounds. Let the nominal lifted block (built with $\hat{A}, \hat{B}_u$) be

$$\mathbf{A}^\star := \mathbf{A}(\hat{A}, \hat{B}_u; X^\star, Y^\star, M^\star, N^\star), \qquad Z^\star_{A,i} := P^\star \mathbf{A}^\star, \quad i \in \{A, B\}.$$

Choose worst-case aligned perturbations that saturate the Young relaxation:

$$E^\star_{A,i} = \begin{cases} \beta_{A,i} \dfrac{Z^\star_{A,i}}{\|Z^\star_{A,i}\|_F}, & Z^\star_{A,i} \neq 0, \\ 0, & Z^\star_{A,i} = 0, \end{cases} \qquad \beta_{A,A} := \beta_A, \quad \beta_{A,B} := \beta_B. \tag{B.7}$$

This choice attains equality in Young's inequality when $s^\star_{A,i} = \|Z^\star_{A,i}\|_F / \beta_{A,i}$.

Step 2: Block structure of $E^\star_A$ and $E^\star_B$. Recall the definitions

$$E_A = \begin{bmatrix} \Delta_A Y^\star & \Delta_A \\ 0 & X^\star \Delta_A \end{bmatrix}, \qquad E_B = \begin{bmatrix} \Delta_B M^\star & \Delta_B N^\star \hat{C}_y \\ 0 & 0 \end{bmatrix}.$$

After choosing $E^\star_A, E^\star_B$ via (B.7), extract the nonzero sub-blocks

$$A_1 := E^\star_A(1{:}n,\ 1{:}n), \quad A_2 := E^\star_A(1{:}n,\ n{+}1{:}2n), \quad A_3 := E^\star_A(n{+}1{:}2n,\ n{+}1{:}2n),$$

$$B_1 := E^\star_B(1{:}n,\ 1{:}m), \qquad B_2 := E^\star_B(1{:}n,\ m{+}1{:}m{+}p),$$

Step 3: Recovery of Δ_A (Sylvester equation). Solve the least-squares normal equation

$$\boxed{(X^{\star\top} X^\star)\,\Delta_A \;+\; \Delta_A\,(Y^\star Y^{\star\top} + I) \;=\; A_2 + A_1 Y^{\star\top} + X^{\star\top} A_3\ ,} \tag{B.8}$$

which is a Sylvester equation in Δ_A. Under mild conditions on the spectra of $X^{\star\top} X^\star$ and $Y^\star Y^{\star\top} + I$, (B.8) has a unique solution.

Special cases. If $X^\star = 0$, then $\Delta_A(Y^\star Y^{\star\top} + I) = A_2 + A_1 Y^{\star\top} \Rightarrow \Delta_A = (A_2 + A_1 Y^{\star\top})(Y^\star Y^{\star\top} + I)^{-1}$. If $Y^\star = 0$, then $(X^{\star\top} X^\star + I)\Delta_A = A_2 + X^{\star\top} A_3 \Rightarrow \Delta_A = (X^{\star\top} X^\star + I)^{-1}(A_2 + X^{\star\top} A_3)$.

Step 4: Recovery of Δ_B (right-side Gram solve). Solve

$$\boxed{\Delta_B\, G_B \;=\; B_1 M^{\star\top} + B_2 \hat{C}_y^\top N^{\star\top}, \qquad G_B := M^\star M^{\star\top} + N^\star \hat{C}_y \hat{C}_y^\top N^{\star\top}\ ,} \tag{B.9}$$

so that, if G_B is invertible, $\Delta_B = (B_1 M^{\star\top} + B_2 \hat{C}_y^\top N^{\star\top})\, G_B^{-1}$. If G_B is singular or ill-conditioned, use the Moore–Penrose pseudoinverse G_B^+ or a Tikhonov-regularized inverse $(G_B + \eta I)^{-1}$ with small $\eta > 0$.

Step 5: Augmented nominal maps. Finally set

$$A_{\text{aug}} = \hat{A} + \Delta_A, \qquad B_{u,\text{aug}} = \hat{B}_u + \Delta_B,$$

and, if desired, verify feasibility with the identification radii: $\|\Delta_A\|_F \le \beta_A$, $\|\Delta_B\|_F \le \beta_B$ and with the lifted bounds $\|E_A(\Delta_A)\|_F \le \beta_A$, $\|E_B(\Delta_B)\|_F \le \beta_B$.

Remarks. (i) The recovered (Δ_A, Δ_B) are *representative* worst-case directions induced by the Young relaxation, not the unique identification errors. (ii) Numerically, (B.8) is solved with a Sylvester solver; (B.9) is a right-multiplication by a Gram factor and should use a robust linear solver or SVD-based pseudoinverse. (iii) If a single uncertainty block on $[A\ B_u]$ was used in synthesis, replace β_A, β_B by a common β and set $E_x^\star = \beta\, Z_x^\star / \|Z_x^\star\|_F$ with $Z_x^\star = P^\star[\hat{A}\ \hat{B}_u]$, then project $E_x^\star$ onto the E_A, E_B structures above before solving (B.8)–(B.9).

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